INCH-POUND

MIL-DTL-30500C
15 April 2025
SUPERSEDING
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DETAIL SPECIFICATION

ROPE, POLYESTER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers polyester rope for general purpose use.

1.2 Classification. The polyester rope may be furnished in one type in the circumferences (size) specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

COMMERCIAL ITEM DESCRIPTION

A-A-59622 - Tags, Shipping and stock

(Copies of these documents are available online at https://assist.dla,mil).

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support – Construction and Equipment (C&E) (ATTN: Code FCSB), 700 Robbins Avenue Philadelphia, PA 19111-5096 or e-mail to CE_Standardization@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ-Z1.4 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of this documents are available online at https://www.asq.org or the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

SAE INTERNATIONAL (SAE)

SAE AMS-STD-595 Colors Used in Government Procurement

(Copies of these documents are available from www.sae.org or SAE International Customer Service, 400 Commonwealth Drive, Warrendale, PA 15096.)

2.4 Order of Precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.1.1.

3.2 Government and contractor purchases. The requirements specified in 3.8, 3.9 and 3.10 apply only to rope purchased directly by the Government. All other requirements apply both to rope purchased by a contractor as a component for an end item and to rope purchased directly by the Government.

3.3 Standard sample. Dyed rope shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6. 3).

3.4 Materials. The material used in the rope shall be bright, virgin, continuous filament polyester fiber of tenacity and denier to meet the requirements of this specification.

3.5 Physical requirements. The rope shall be of three strands and shall conform to the requirements specified in table I when tested as specified in 4.2.4. The lay of the rope shall be right hand or “Z” lay. Each strand shall be made of one size of one single polyester yarns or one size of balanced 3 ply polyester yarns in sizes 0.625 inch to 1 inch circumference inclusive. For rope sizes from 1.125 inch circumference and up, each strand shall be made of one size of balanced 3 ply polyester yarns.

3.5.1 Elongation. The elongation of the rope shall not exceed 35 percent when tested as specified in 4.2.4.

3.5.2 Hardness. The rope shall meet the minimum and shall not exceed the maximum hardness requirements specified in table I, when tested as specified in 4.2.4.

3.6 Finish. No extraneous material shall be added for the purpose of weighting the rope. The extractable matter of the finished rope shall not exceed 4.0 percent when as specified in 4.2.4.

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TABLE I. Physical requirements.

Circumference	Circumference tolerance of plus or minus	Diameter 1/ (approx..)	Length per pound 2/ (min.)	Breaking strength (min.)	Length per reel (min.)	Weight per reel (approx.) 1/ 3/	Hardness (min.)	Hardness (max.)
Inches	Inches	Inches	Feet	Pounds	Feet	Pounds	Pounds	Pounds
0.625	0.063	0.188	84.0	800	2,250	28	5	25
0.750	0.063	0.250	50.0	1,200	2,250	48	5	25
1	0.063	0.313	33.0	2,000	2,250	72	5	25
1.125	0.063	0.375	23.5	2,800	1,600	73	5	25
1.250	0.063	0.438	17.0	3,800	1,200	74	5	25
1.500	0.063	0.500	13.0	5,000	1,200	97	5	25
1.750	0.063	0.563	10.0	6,500	1,200	126	5	25
2	0.125	0.625	8.0	8,000	1,200	158	5	25
2.250	0.125	0.750	6.0	10,000	1,200	210	5	25
2.500	0.125	0.813	5.3	13,000	1,200	238	5	25
2.750	0.125	0.875	4.2	15,000	1,200	300	5	25
3	0.125	1	3.5	18,500	1,200	360	20	100
3.500	0.188	1.125	2.5	25,000	1,200	504	20	100
4	0.188	1.313	2.0	31,000	600	315	20	100
5	0.250	1.625	1.30	48,000	600	485	20	100
6	0.313	2	0.85	68,000	600	742	20	100
7	0.375	2.250	0.66	88,000	600	956	20	100
8	0.375	2.625	0.50	110,000	600	1,260	20	100
9	0.438	3	0.40	140,000	600	1,575	20	100
10	0.500	3.250	0.33	165,000	600	1,910	20	100
11	0.563	3.500	0.25	240,000	600	2,400	20	100
12	0.625	4	0.20	285,000	600	2,800	20	100

1/ Not a specification requirement – included for information purposes only.

2/ A minus tolerance of 10 percent is allowed for dyed rope.

3/ A plus tolerance of 10 percent shall be allowed for dyed rope.

3.7 Color. Unless otherwise specified (see 6.2), the color of the finished rope shall be natural. When colored rope is specified the rope shall be dyed Green 383 Camo, to approximate color chip 34094 of SAE AMS-STD-595 (see 6.3 and 6.6).

3.7.1 Colorfastness. Dyed and finished rope shall show fastness to weathering equal to or better than the standard sample when tested as specified in 4.2.4. When no standard sample has been established, dyed rope shall show good fastness to weathering when tested as specified in 4.2.4.

3.8 Identification. Each reel (spool) of rope shall have a ticket (identification tag) or label attached for identification purposes. The ticket shall be in accordance with A-A-59622. The label shall be attached in such a manner as to remain in place and be clearly legible until all rope has been removed. All ticket or label entries shall be legibly printed, stamped, or typed with water insoluble ink. The ticket or label shall contain the following information:

National stock number
Nomenclature
Specification number
Length
Size (circumference)
Contract number and date
Contractor's name
Date of manufacture (month and year)

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3.8.1 Identification marker. The manufacturer shall identify his product by inserting a kraft paper or water repellent cotton marker within one strand in all ropes larger than 1.125 inch circumference. The marker shall be completely enveloped by the cover yarns in the strand in which it is enclosed. Unless otherwise specified, the manufacturer's name, year of manufacture and type of fiber (POLYESTER) shall be clearly printed in block type on the marker. The printing shall not be affected upon exposure to water or mineral oil.

3.9 Put-up. Unless otherwise specified, (see 6.2) the rope shall be furnished without knots or splices on non-returnable reels (spools) not larger than 6 feet in diameter in the lengths specified in table I. Pieced lengths will be permitted, but no piece shall be less than 600 feet when measured in relaxed condition. The ends of all rope shall be cut off squarely and be securely whipped, taped or heat sealed to prevent untwisting or fraying. The reels (spools) shall be wound so that each turn and layer is free from entanglement.

3.9.1 When specified, rope shall be put up in coils in one continuous piece without knots or splices in the minimum length specified (see 6.2). A plus tolerance of 10 percent shall be allowed on the length per coil. The ends of all rope shall be cut off squarely and be securely whipped, taped or heat sealed to prevent untwisting or fraying. The coils shall be so wound that each turn and layer is free from entanglement.

3.10 Moisture content. The moisture content of the rope, as received, shall not exceed 2.0 percent when tested as specified in 4.2.4 (see 6.5).

3.11 Spliceability. The finished rope shall be spliceable and shall not develop yarn displacement or strand cockles in the splicing test specified in 4.2.4.

3.12 Workmanship. The rope shall conform to the quality and grade of product established by this specification.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1).
b. Conformance inspection (see 4.2).

4.1.1 First article inspection. When required (see 6.2), the preproduction sample submitted in accordance with 3.1 shall be visually inspected for appearance and color. The sample shall be tested for physical and chemical characteristics in accordance with 4.2.4.

4.2 Conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with ASQ-Z1.4.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents. In addition, inspection shall be performed for the requirements in table II and a supplier's certificate of compliance thereto shall be furnished and will be accepted for the requirements.

Table II. Component test.

Characteristic	Paragraph reference
Material	3.4
Yarn size	3.5
Single yarn in strand (when applicable)	3.5
Balanced plied yarn in strand (when applicable)	3.5

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4.2.2 Examination of the end item for visual defects. The defects listed in table III shall be counted regardless of their proximity to each other. The sample unit shall be one reel (spool) or coil as applicable. Ten percent of the length contained on each sample unit, but not less than 100 feet shall be subjected to the visual examination. The lot size for this examination shall be expressed in units of reels (spools) or coils each. The inspection level shall be level I.

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Table III. Visual examination defects.

Examine	Defect	Classification	
		Major	Minor
Appearance and workmanship	Cut, any	X	
	Chafed or damaged	X	
	Kink, darting yarn 1/, broken or loose end, bulged strand, strand knot	X	
	Ends not cut off squarely		X
	Ends not securely whipped, taped or heat sealed		X
	Splice or otherwise secured to make a continuous length	X	
Identification	Omitted, incorrect, or illegible		X
	Insecurely attached or not as specified		X
Color	Not as specified	X	
	Resists and/or spotty dyeing		X
	Specific shade not within established tolerances		X
	Uneven		X
Cleanness	Spot, or stain, clearly noticeable 2/		X

1/ Darting yarns are internal yarns which projects through the cover yarns of the strand.

2/ At normal inspection distance (approximately 3 feet).

4.2.3 Examination for length and winding. The sample unit for this examination shall be one reel (spool) or coil. The inspection level shall be level S-3. The minimum sample size shall be code letter “E” (sample size 13). The lot size shall be the number of units in the inspection lot. Defects shall be as listed in 4.2.3.1 and 4.2.3.2. The required length of the spool (reel) or coil (see table I and 3.9.1) shall be determined using the following formulas:

W x L = Length per holder in feet

Where: W is net weight of the rope in 0.1 pounds

L is the lot average of the weight-length relationship (length/pound) as determined in 4.2.4

4.2.3.1 Defects with regard to length shall be considered to exist if any of the following are determined during inspection:

a. Length of spool (reel) less than specified.

b. Length of coil less than or more than length specified (including permitted tolerances).

c. Length of unit less than marked on ticket.

d. Rope not in continuous length when specified (see 3.9.1).

e. Any piece on reels (spools) less than 600 feet in length.

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4.2.3.2 Defects with regard to winding shall be considered to exist if any of the following are determined during inspection:

a. Improperly or not firmly wound resulting in kinks, knots, entangling or slippage during unwinding or otherwise affecting free unhampered unwinding of rope.

b. Knot or splice or otherwise joining of ends to make a continuous length.

4.2.4 Testing of the end use. The methods of testing specified in FED-STD-191, wherever applicable and as listed in table IV shall be followed. When data in the "Number of determinations" and "Results reported as" columns are not specified in the table, they shall be as required by the referenced test method. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample size shall be in accordance with table V. The sample unit for testing shall be one reel (spool) or coil of rope. The lot size shall be expressed in units of one reel (spool) or coli. All test reports shall contain the individual values utilized in expressing the final result. Tests to determine compliance with specification requirements, including quantity of delivery, may be made under prevailing atmospheric conditions. In case of dispute, tests shall be made upon material which has reached equilibrium under Standard Conditions as defined in FED-STD-191.

TABLE IV. Test methods.

Characteristic	Requirement reference	Test method
Number of strands	3.5	Visual 1/
Lay of rope	3.5	Visual 1/
Circumference	3.5	6003
Length per pound	3.5	6004
Breaking strength	3.5	6015.1 2/
Elongation	3.5.1	6015.1 2/
Hardness	3.5.2	6020
Extractable matter	3.6	2611 3/
Color (shade) matching	3.7	4.2.4.1 1/
Colorfastness to weathering	3.7.1	5671.1 4/
Identification marker	3.8.1	Visual 5/
Moisture content	3.10	2600
Spliceability	3.11	4.2.4.2 1/

1/ One determination per sample unit shall be made and the results shall be reported as pass or fail.

2/ Method 6016 of FED-STD-191 may be used to determine breaking strength and elongation for ropes 3 inches in circumference or less.

3/ Except that extractable matter shall be determined using the chloroform extraction procedure.

4/ The time of exposure shall be 40 standard fading hours.

5/ Supplier's certificate of compliance thereto shall be furnished and will be accepted for the requirement.

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TABLE V. Sampling for tests.

Number of Reels (spools) or coils in lot	Number of samples
800 or less	2
801 up to and including 22,000	3
22,000 and over	5

4.2.4.1 Determination of shade (color) matching. The shade of the dyed rope shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be good approximation to the standard sample under incandescent lamplight at 2800° Kelvin (see 6.3).

4.2.4.2 Determination of spliceability. A breaking strength specimen shall be prepared and spliced at each end with four full tucks. The specimen shall then be allowed to rest for 24 hours. Three of the tucks shall then be backed out of the rope. That portion of the rope from which the tucks are removed shall be examined for yarn displacement and strand cockles.

5. PACKAGING

5.1. Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Rope covered by this specification is intended for general purpose use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Circumference (size) required (see table I).
c. First article, when first article sample is required (see 3.1, 4.2 and 6.4).
d. Color required, if other than natural (see 3.7).
e. Put-up and length when other than specified (see 3.9).
f. Packaging requirements (see 5.1).
g. That polyester rope may be purchased on a price-per-pound basis net weight.
h. That purchaser will accept at original weight, any unit which has been shortened or cut for test specimens, if in compliance with this specification.

6.3 Standard sample. For access to the standard sample, address the contracting activity issuing the invitation for bids or request for proposal.

6.4 First article. When a first article is required, it should be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.5 Subject term (key word) listing.

Continuous filament polyester fiber
Reel
Spool

6.6 Color. Olive green and olive drab colors have been replaced by Camouflage Green 383. Any item which previously required any olive green or olive drab color will now require Camouflage Green 383.

6.7 Adjustment for high moisture content. Material furnished containing an excess of moisture will be accepted by an adjustment in weight to the 2 percent moisture basis.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - GL
Navy - SH
Air Force - 71
DLA - CQ

Review activities:
Army - CR4

Preparing Activity:
DLA - CQ

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